

Real Numbers

The Fundamental Theorem of Arithmetic and proofs of irrationality, with worked examples and practice questions from the CBSE 2026-27 syllabus.

Introduction

Real numbers include rational numbers, which can be written as p/q where p and q are integers and $q \neq 0$, and irrational numbers, which cannot. For CBSE 2026-27, this chapter focuses on the Fundamental Theorem of Arithmetic and algebraic proofs that numbers such as $\sqrt{2}$, $\sqrt{3}$ and $\sqrt{5}$ are irrational.

The Fundamental Theorem of Arithmetic

You already know that any composite number can be broken down into a product of primes (for example, $30 = 2 \times 3 \times 5$). This chapter states that fact formally, and adds an important detail: this factorisation is unique.

Fundamental Theorem of Arithmetic

Every composite number can be expressed (factorised) as a product of primes, and this factorisation is unique, apart from the order in which the prime factors occur.

Worked Example

Express 140 as a product of its prime factors.

$$140 = 2 \times 70 = 2 \times 2 \times 35 = 2 \times 2 \times 5 \times 7 = 2^2 \times 5 \times 7$$

No matter which pair of factors you start with, you will always arrive at the same set of prime factors. This is what "unique factorisation" means.

Proving a Number is Irrational

A number is **irrational** if it cannot be written in the form p/q , where p and q are integers and $q \neq 0$. In this section, we prove that certain square roots are irrational. This is one of the most important proof-based questions in this chapter, and it appears frequently in board exams.

Theorem

$\sqrt{2}$ is irrational.

Proof. We prove this by contradiction: we assume the opposite of what we want to prove, and show that this assumption leads to something impossible.

Let us assume, to the contrary, that $\sqrt{2}$ is rational. Then we can write

$$\sqrt{2} = p/q$$

where p and q are coprime integers (that is, $\text{HCF}(p, q) = 1$) and $q \neq 0$. This is always possible, since any rational number can be reduced to its simplest form.

Squaring both sides:

$$2 = p^2/q^2 \implies p^2 = 2q^2 \quad \dots(1)$$

From (1), p^2 is divisible by 2. It follows that p itself must be divisible by 2: if a prime number divides the square of an integer, it must also divide the integer itself, by the Fundamental Theorem of Arithmetic. So we can write $p = 2c$ for some integer c .

Substituting $p = 2c$ back into (1):

$$(2c)^2 = 2q^2 \implies 4c^2 = 2q^2 \implies q^2 = 2c^2$$

This shows q^2 is also divisible by 2, and so q must be divisible by 2 as well.

But this means both p and q are divisible by 2, which contradicts our assumption that p and q are coprime ($\text{HCF}(p, q) = 1$). This contradiction has arisen because of our incorrect assumption that $\sqrt{2}$ is rational.

So, $\sqrt{2}$ is irrational.

The same method (assume the number is rational, write it in the form p/q with p and q coprime, and show that this forces both p and q to share a common factor) can be used to prove that $\sqrt{3}$, $\sqrt{5}$, and in general $\sqrt{p}$ for any prime number p , are irrational.

Worked Example

Prove that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational.

Assume, to the contrary, that $3 + 2\sqrt{5}$ is rational. Then we can write

$$3 + 2\sqrt{5} = p/q$$

for some coprime integers p and q , with $q \neq 0$. Rearranging,

$$2\sqrt{5} = p/q - 3 = (p - 3q)/q \implies \sqrt{5} = (p - 3q)/2q$$

Since p and q are integers, the right-hand side $(p - 3q)/2q$ is a rational number. This would mean $\sqrt{5}$ is rational, but this contradicts the known fact that $\sqrt{5}$ is irrational.

This contradiction shows our assumption was wrong. So, **$3 + 2\sqrt{5}$ is irrational.**

This same reasoning applies generally: the sum of a rational number and an irrational number is always irrational, and the product of a nonzero rational number and an irrational number is always irrational.

Summary

- **Fundamental Theorem of Arithmetic:** every composite number can be factorised into primes in exactly one way, apart from the order of the factors.
- $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, and $\sqrt{p}$ for any prime p , are irrational (proved by contradiction).
- The sum of a rational number and an irrational number is irrational. The product of a non-zero rational number and an irrational number is also irrational.

Practice Questions

1. Express 156 as a product of its prime factors.

2. Prove that $\sqrt{3}$ is irrational.

3. Prove that $5 - \sqrt{2}$ is irrational.